PRESENTATION SCRIPT — TWO SPEAKERS

Development of an Image-based Disease Detection System Leveraging AI/ML Techniques and Computer Vision

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Edited for a natural spoken delivery: clear, conversational, and technically precise. Numerical values and the slide order have been retained.

Speaker split: Aakash introduces the project and explains the data/model work; Kamaaksha covers the timeline, evaluation, and publication. Both speakers close together.

SLIDE 1 — TITLE

Speaker: AAKASH

Good morning/afternoon, everyone. I'm Aakash Jawle, and I'm here with my co-fellow, Kamaaksha Kharul. We'll be presenting the work we carried out through the CHANAKYA Fellowship at TIH, IIT Bombay. Our project was titled "Development of an Image-based Disease Detection System Leveraging AI/ML Techniques and Computer Vision," and we worked under the mentorship of Dr. Santwana Gudadhe at Pimpri Chinchwad College of Engineering, Pune.

Our goal was to explore how AI can help identify diseases and pests from crop images. We worked with tomato, mango, and soybean, and tried both conventional CNN models and newer vision-language models. Along the way, we built data-checking and fine-tuning workflows, and tested not just how well the models did on familiar images, but also how they handled images from a different source. We'll walk you through what we tried, what worked, and what still needs improvement.

SLIDE 2 — PROJECT CARD

Speaker: AAKASH

The fellowship ran from 1 December 2025 to 30 September 2026. The total sanctioned amount was ■2.40 lakh, and TIH disbursed ■1.92 lakh. That works out to ■1.20 lakh sanctioned per fellow and ■0.96 lakh disbursed per fellow, with ■0.48 lakh still pending.

One major outcome was that our paper was accepted for the Proceedings of ICAIN 2026 in Springer's LNNS series, under Submission ID 953. The acceptance notification placed it among the top 20% of papers reviewed. We did not file any IP or patents, and we did not commercialise the work during the fellowship. What we have produced is a set of research workflows, datasets, and trained adapters. Before this can be used in the field, it still needs to be tested on independently collected field images.

SLIDE 3 — BACKGROUND & PROBLEM STATEMENT

Speaker: AAKASH

Crop diseases and pests can look quite similar in photographs, so identifying them reliably is not always straightforward. Manual inspection can take time and often depends on access to an experienced person. We focused on three crops—tomato, mango, and soybean—using separate experimental tracks: TIH and PlantVillage data for tomato, PlantDx and MangoLeafBD for mango-related work, and SoyNet alongside the planned TIH-VDI track for soybean.

Our starting point was a general-purpose vision-language model, without task-specific fine-tuning. In this zero-shot baseline, accuracy was 7.3% for tomato and 17.5% for mango. Those results gave us a clear reason to explore CNN baselines and parameter-efficient fine-tuning. We also wanted to answer a more practical question: would a model that performs well on its own test data still work on images collected in a different setting?

SLIDE 4 — OBJECTIVES

Speaker: AAKASH

We set eight objectives. First, compare several vision-language model families and choose a suitable backbone. Second, check the datasets carefully for broken images, duplicates, and label conflicts, then create fixed training, validation, and test splits. Third, fine-tune Qwen2.5-VL-7B-Instruct with QLoRA on a single 24 GB GPU, instead of updating every parameter in the model.

The remaining objectives were to build an 11-class tomato classifier and an 8-class mango classifier; test generalisation on an independent external benchmark; and create a knowledge-grounding layer with cited disease information, an evidence-linked ontology, a controlled vocabulary, and a symptom lexicon. Finally, we wanted the work to be reproducible, so we used fixed seeds, configuration-driven workflows, validation checks, tests, and documented evaluation results.

SLIDE 5 — EXPECTED BENEFITS OF THE PROJECT

Speaker: AAKASH

The broader aim is to make image-based disease and pest identification easier to investigate and potentially support, rather than replace, expert judgement. One technical result is that we fine-tuned 40.37 million LoRA parameters on a single 24 GB GPU, with peak memory use of 10.57 GB, instead of fine-tuning the full 7-billion-parameter model.

We also removed 550 duplicate or contradictory files and froze the final data splits, so images would not quietly move between training, validation, and testing. Fixed seeds, split hashes, preflight checks, and tests help make the work repeatable. Comparing internal and external results also helped us identify where the model struggles, especially on Late Blight. The approach could be extended to other crops, but more field data and independent validation are needed before real-world use.

I'll now hand over to Kamaaksha, who will take us through the timeline and the evaluation results.

SLIDE 6 — OVERALL WORK PROGRESS & MILESTONE TIMELINE

Speaker: KAMAAKSHA

Thanks, Aakash. We moved through eight main milestones during the fellowship. In December 2025 and January 2026, we built the foundations: image preprocessing, quality checks, augmentation, and CNN baselines. We trained a custom CNN from scratch and tested EfficientNetB3, MobileNetV2, and ConvNeXt-Tiny. The custom CNN reached 98.51% on tomato.

In January and February, we extended the experiments to soybean, including ConvNeXt-Tiny in Keras and PyTorch, SoyNet CNN experiments, and the TIH-VDI track. ConvNeXt-Tiny in PyTorch reached 99.92%, while SoyNet ResNet50 reached 95.16%. From March to June, we compared vision-language models and selected Qwen2.5-VL-7B-Instruct, which scored 98.50% in the screening benchmark. Between May and July, we developed PlantDx and the knowledge-grounding layer, including a disease knowledge base, ontology, and a 1,966-caption corpus.

In July, the knowledge-grounded QLoRA pipeline raised tomato accuracy from 7.3% to 93.7% and mango accuracy from 17.5% to 82.0%. In August and September, we audited the final TIH tomato dataset, reducing it from 7,095 to 6,545 images, and trained the final Qwen2.5-VL-7B model using QLoRA. It achieved 92.76% accuracy on the internal held-out test and 69.32% on an independent external test of 176 images. In October 2026, the paper was accepted for Springer's LNNS series. The fellowship itself ended on 30 September 2026.

SLIDE 7 — PROJECT OVERVIEW & TECHNICAL PROGRESSION

Speaker: KAMAAKSHA

If we look at the project as a whole, it moved through ten stages: foundational work, CNN baselines, VLM selection, knowledge-grounded diagnosis, TIH dataset adaptation, QLoRA fine-tuning, validation, internal testing, cross-domain evaluation, and publication. In simple terms, we first built and checked the pipeline, then trained the models, and finally tested how reliable the results were.

The central challenge was to classify crop diseases and pests from images, even though symptoms can look similar and manual inspection can be slow. We combined CNNs and vision-language models, using CUDA-based

GPU workflows and Apple Silicon with MLX for selected experiments. We treated external evaluation as essential, because a strong score on familiar images does not automatically mean the model will perform well on new ones.

Across the three crop tracks, the final TIH tomato model reached 92.76% held-out accuracy and 69.32% external accuracy. The knowledge-grounded mango model improved from 17.5% zero-shot to 82.0% after QLoRA fine-tuning, and the SoyNet soybean CNN track reached 95.16% with ResNet50. Our outputs include 7 GitHub repositories under Apache-2.0, trained CNN checkpoints and QLoRA adapters, audited datasets with frozen splits, evaluation reports and confusion matrices, a Streamlit research prototype, and the accepted ICAIN 2026 paper.

SLIDE 8 — TOMATO: CNN BASELINES (PLANTVILLAGE)

Speaker: AAKASH

We started the tomato CNN experiments with PlantVillage and trained 5 architectures. ConvNeXt-Tiny in PyTorch reached 99.92% accuracy using a disjoint 70/15/15 split with 2,411 test images. This is the stronger benchmark to highlight because the test images were kept separate from the training data.

Our custom CNN, trained from scratch, reached 98.51%. However, that result is only indicative because its 80/20 split overlapped substantially with the training data. ConvNeXt-Tiny in Keras reached 96.15% on a held-out validation set of 3,198 images. We used 224×224 input images and an RTX 4500 Ada GPU. The PyTorch ConvNeXt-Tiny result also had a Macro-F1 of 0.9991.

We also added a check before prediction to help reject inputs that clearly do not look like crop images. It uses the HSV green-pixel ratio, edge ratio, and cosine similarity to MobileNetV2 embeddings. The takeaway here is that the score alone is not the whole story: how the data is split matters. A result from a split with overlap should not be compared directly with one from a fully separate test set.

SLIDE 9 — SOYBEAN: CNN CLASSIFICATION (SOYNET & TIH VDI)

Speaker: AAKASH

For soybean, our SoyNet dataset contained 3,720 camera-click images: 3,164 diseased and 556 healthy. We resized the images to 256×256 and applied augmentation only to the training set. Using a stratified 70/15/15 split with seed 42 gave us 2,603 training images, 557 validation images, and 560 test images.

We tested 6 backbones across Keras and PyTorch: ResNet50 and MobileNetV2 in both frameworks, plus EfficientNetB0 and EfficientNetB3 in PyTorch. The strongest result was ResNet50 in Keras, at 95.16% accuracy and ROC-AUC 0.9903. MobileNetV2 in Keras reached 89.07%. In the PyTorch runs, ResNet50, MobileNetV2, EfficientNetB0, and EfficientNetB3 achieved recall of 1.0; EfficientNetB3 had the strongest PyTorch accuracy at 93.93%.

There is one important limitation: we could not include results from the separate TIH-VDI soybean model because the VDI environment was unavailable. The dataset is also imbalanced, with approximately 5.7 diseased images for every healthy image, so minority-class precision, recall, and F1 need to be interpreted carefully.

I'll hand back to Kamaaksha for the vision-language model selection.

SLIDE 10 — TOMATO: VLM MODEL SELECTION

Speaker: KAMAAKSHA

Before choosing our VLM backbone, we screened several candidates using the same PlantVillage structured-caption protocol. LLaVA-1.5 and LLaVA-1.6 were discontinued because of batch-collation incompatibility. BLIP-2 with OPT-2.7B was able to produce the required structured output and reached 89.00% accuracy. We also tried Florence-2, but compatibility problems in April and again in June meant we had to stop that track.

We selected Qwen2.5-VL-7B-Instruct as our backbone in April. It reached 98.50% accuracy in the screening benchmark. The chart shows 2% for LLaVA, effectively 0% for Florence-2, 89% for BLIP-2, and 98.5% for Qwen2.5-VL. We used the same protocol to make the comparison as consistent as possible, although the models that failed to produce usable outputs could not be carried forward.

Qwen2.5-VL also produced zero unparseable outputs in the reported runs. That mattered because we needed predictions that could be evaluated consistently, not just a high accuracy score. Its screening performance and

output reliability made it the backbone for the later PlantDx and final TIH tomato experiments.

SLIDE 11 — PLANTDX & TOMATO PIPELINE

Speaker: KAMAAKSHA

In May 2026, we completed an intermediate milestone called PlantDx. We set up a 15-class task and fine-tuned Qwen2.5-VL using LoRA, with run ID run_001, on Apple Silicon through MLX.

On a frozen held-out split of 200 PlantVillage images, this run achieved 98.00% accuracy and Macro-F1 0.9777. We also built a Streamlit inference app, version 1.0.0, so we could compare the base model's predictions with the fine-tuned model on that fixed test split. We used the app for the review demonstration.

To be clear, the Streamlit app was a research prototype, not a production deployment. The 15-class PlantDx result was an intermediate milestone. It came before the larger knowledge-grounding work and the final TIH tomato training, which used a separately audited dataset and a different evaluation protocol.

Back to Aakash for the knowledge-grounded tomato and mango pipeline.

SLIDE 12 — KNOWLEDGE-GROUNDED TOMATO & MANGO PIPELINE

Speaker: AAKASH

At this stage, we added a structured knowledge-grounding layer. We built a disease knowledge base covering 18 cited conditions—10 for tomato and 8 for mango—and organised it into a typed, evidence-linked ontology with a controlled vocabulary and symptom lexicon.

We generated the caption corpus deterministically from concepts and templates, then validated it through multiple checks. Importantly, we did not use a VLM, LLM, or image analysis to generate these captions; they came from cited disease knowledge. We applied the workflow to audited, normalised PlantVillage tomato images and MangoLeafBD mango images, then fine-tuned Qwen2.5-VL-7B using 4-bit QLoRA with rank 16 on Apple Silicon.

For tomato, accuracy rose from 7.3% zero-shot to 93.7% on 910 test images, with Macro-F1 0.919 and Weighted-F1 0.937. For mango, accuracy rose from 17.5% to 82.0% on 200 test images, with Macro-F1 and Weighted-F1 both at 0.811. We focused on using cited knowledge, describing symptoms and severity carefully, and keeping the workflow reproducible.

One caveat is worth keeping in mind: these results came from public, single-leaf datasets and are in-distribution results. They are separate from the final TIH benchmark, so the numbers should not be compared directly.

SLIDE 13 — FINAL TIH TOMATO DATASET & TRAINING

Speaker: AAKASH

For the final TIH tomato experiment, we started with 7,095 labelled images across 11 class folders. We checked file formats and whether each image could be opened correctly. Then we used SHA-256 exact deduplication, removing repeated images within the same class and excluding contradictory duplicates where the same hash appeared under more than one label. That left 6,545 images, or 92.25% of the original dataset.

We created a class-stratified 80/10/10 split using seed 20260911: 5,247 images for training, 649 for validation, and 649 for testing. We froze the split hashes so the partitions would stay fixed. The classes were imbalanced: Late Blight had 195 test images, while Healthy Fruit had 9. To reduce the effect of that imbalance during training, we used inverse-square-root sampling.

Our base model was Qwen2.5-VL-7B-Instruct, with the vision encoder frozen. We used 4-bit NF4 quantisation, double quantisation, and BF16 compute. The LoRA adapters used rank 64, alpha 128, and dropout 0.05, and were applied to the query, key, value, and output projections. This gave us 40.37 M trainable parameters. We used Paged AdamW 8-bit, a learning rate of 4e-5, cosine scheduling, and 3% warm-up. With batch size 1 and 8 gradient-accumulation steps, the effective batch size was 8. We planned 3 epochs and completed 2.909. For checkpoint selection, we prioritised validation Macro-F1, then accuracy, then loss, with early stopping patience 1.

I'll hand over to Kamaaksha to explain how the model performed on the held-out test set.

SLIDE 14 — EXPERIMENTAL RESULTS: INTERNAL TEST (TIH, 649 IMAGES)

Speaker: KAMAAKSHA

On the held-out TIH test set of 649 images, the final model achieved 92.76% accuracy, correctly classifying 602 of 649 images. Macro-F1 was 93.78%, which we used as our main summary metric because the classes were imbalanced. Macro recall was 95.57%, macro precision was 92.71%, and inference took 0.327 seconds per image on a single RTX 4500 Ada GPU.

Training involved 1,908 optimizer steps across 2.909 epochs and took 192.7 minutes. Peak memory use was 10.57 GB. The model produced zero unparseable outputs across the test set, so every output could be processed by the evaluation pipeline.

Looking at the individual classes gives us a more useful picture than accuracy alone. Late Blight, which had 195 test images, achieved F1 90.91% and recall 84.62%; its main confusions were Target Leaf Spot, Leaf Curl, and Leaf Miner. Target Leaf Spot had 100% recall but 68.75% precision, giving an F1 of 81.48%. Healthy Fruit scored 100%, but there were only 9 test images in that class, so we should be cautious about drawing strong conclusions from it. These examples show why class-level results matter.

SLIDE 15 — CROSS-DOMAIN EVALUATION (INTERNAL VS EXTERNAL)

Speaker: KAMAAKSHA

This was one of the most important checks in the project: we wanted to see how the same model handled images from outside its internal test distribution. We evaluated it on an independent external benchmark of 176 images across 10 of the 11 classes.

The difference was clear. Internal accuracy was 92.76%, while external accuracy was 69.32%, a drop of 23.44 percentage points. Macro-F1 went from 93.78% to 71.95%, down 21.83 points. Inference latency also increased from 0.327 to 0.373 seconds per image.

The change varied by class. Late Blight F1 fell from 90.91% internally to 14.29% externally. Of the 20 external Late Blight images, 11 were misclassified as Target Leaf Spot. Mealy Bugs and White Fly were more consistent, with scores of 93.33% and 91.67%, respectively.

So the external benchmark revealed a substantial domain gap. Strong performance on the internal test set does not, on its own, show that the model is ready for field use. We would need broader and more varied training data, followed by more testing on independently collected images.

SLIDE 16 — RESULTS AT A GLANCE: ALL EXPERIMENTS

Speaker: AAKASH

This slide brings the experiments together, but it is not a single leaderboard. The results come from four datasets—TIH tomato, PlantVillage, MangoLeafBD, and SoyNet—and different methods, including zero-shot inference, LoRA, QLoRA, and standard CNN transfer learning. We also used different hardware setups: CUDA on the RTX 4500 Ada and Apple Silicon through MLX.

The custom CNN reached 98.51% on PlantVillage, although its split overlapped with training, so that result is indicative only. ConvNeXt-Tiny in PyTorch reached 99.92% on a disjoint split. Qwen2.5-VL screening reached 98.50%, and the intermediate 15-class PlantDx run reached 98.00%. The knowledge-grounded QLoRA pipeline reached 93.7% on tomato and 82.0% on mango. SoyNet ResNet50 reached 95.16% on soybean. Finally, the TIH tomato model achieved 92.76% on the internal held-out set and 69.32% on the external benchmark.

It is important to compare these numbers within their own datasets and evaluation protocols. PlantVillage images are not the same as the TIH tomato images, and averaging these accuracies would be misleading. We compare each zero-shot result with its own fine-tuned result. The main takeaway is the drop from 92.76% internal accuracy to 69.32% external accuracy, which shows that cross-domain robustness still needs work.

SLIDE 17 — PUBLICATION, MILESTONE STATUS & LIMITATIONS

Speaker: KAMAAKSHA

Our paper, “Efficient QLoRA Fine-Tuning of Qwen2.5-VL for Multi-Class Tomato Disease Classification and Cross-Domain Evaluation,” has been accepted at ICAIN 2026 for Springer’s LNNS series under Submission ID 953. The authors are Aakash Trimbak Jawle, Kamaaksha Rajendra Kharul, Dr. Santwana Gudadhe, Harish Lokawad, and Rutuja Chavan. Registration was completed on 20 October 2026. The acceptance notification placed the paper among the top 20% of papers reviewed. DOI, volume, and page numbers have not yet been assigned.

We completed the CNN and VLM baselines, QLoRA fine-tuning, and internal and external evaluation across the three crop tracks. There are still some limitations. The PlantVLM caption-retrieval experiment was exploratory and tested on only 30 images. We stopped the Florence-2 track because of compatibility issues. The separate TIH-VDI soybean model could not be included because the VDI environment was unavailable. Most importantly, external accuracy was 23.44 percentage points below internal accuracy, showing that the model still faces a domain gap.

To summarise the project status, ■2.40 lakh was sanctioned, ■1.92 lakh was disbursed, and ■0.48 lakh remains pending. We did not file any IP or patents. The fellowship ran from 1 December 2025 to 30 September 2026.

SLIDE 18 — CONCLUSION & FUTURE WORK

Speaker: KAMAAKSHA, THEN AAKASH

To wrap up, we built and evaluated CNN and vision-language workflows for disease and pest classification across tomato, mango, and soybean. We moved from image checks and dataset auditing to CNN baselines, knowledge grounding, QLoRA fine-tuning, and then internal and cross-domain evaluation. Our key results were 92.76% internal accuracy and 93.78% Macro-F1 on TIH tomato, 93.7% accuracy on knowledge-grounded tomato, 82.0% on mango, and 95.16% on SoyNet soybean. Our tomato research has been accepted at ICAIN 2026.

At the same time, we want to be clear about the main limitation: external accuracy was 69.32%, and Late Blight F1 dropped to 14.29% on the external benchmark. High performance on familiar images is encouraging, but it is not enough to claim reliable real-world performance.

Our next steps are to test the system on independently collected field images, expand the datasets and crop coverage, improve inference and deployment, and strengthen the knowledge-grounding layer. Any decision about deployment should come after that validation—not just from the internal test score.

BOTH: Thank you.

SLIDE 19 — THANK YOU

Speaker: BOTH

Thank you for your time and attention. We’re happy to answer your questions.